Task 12: Jenkins Docker Continuous Deployment

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1. Objective

Integrate Docker image creation and container deployment into Jenkins so a successful build automatically produces and deploys the application container.

2. Jenkinsfile

```
pipeline {
    agent any

    stages {
        stage('Checkout') {
            steps {
                checkout scm
            }
        }

        stage('Build and Test') {
            steps {
                bat '"C:\DevTools\apache-maven-3.9.16\bin\mvn.cmd" clean test package'
            }
        }

        stage('Docker Build') {
            steps {
                bat 'docker build -t delivery-promise-calculator:1.0 .'
            }
        }

        stage('Docker Deploy') {
            steps {
                bat '''
                docker rm -f delivery-promise-app 2>nul || exit /b 0
                docker run -d --name delivery-promise-app -p 8766:8765 delivery-promise-calculator:1.0
                '''
            }
        }

        stage('Verify Deployment') {
            steps {
                bat '''
                docker ps --filter "name=delivery-promise-app"
                echo Docker deployment completed successfully.
                '''
            }
        }
    }

    post {
        success {
            echo 'Pipeline completed successfully. Docker application deployed on port 8766.'
        }
        failure {
            echo 'Pipeline failed. Check the stage logs.'
        }
    }
}
```

3. Dockerfile Checkout Problem

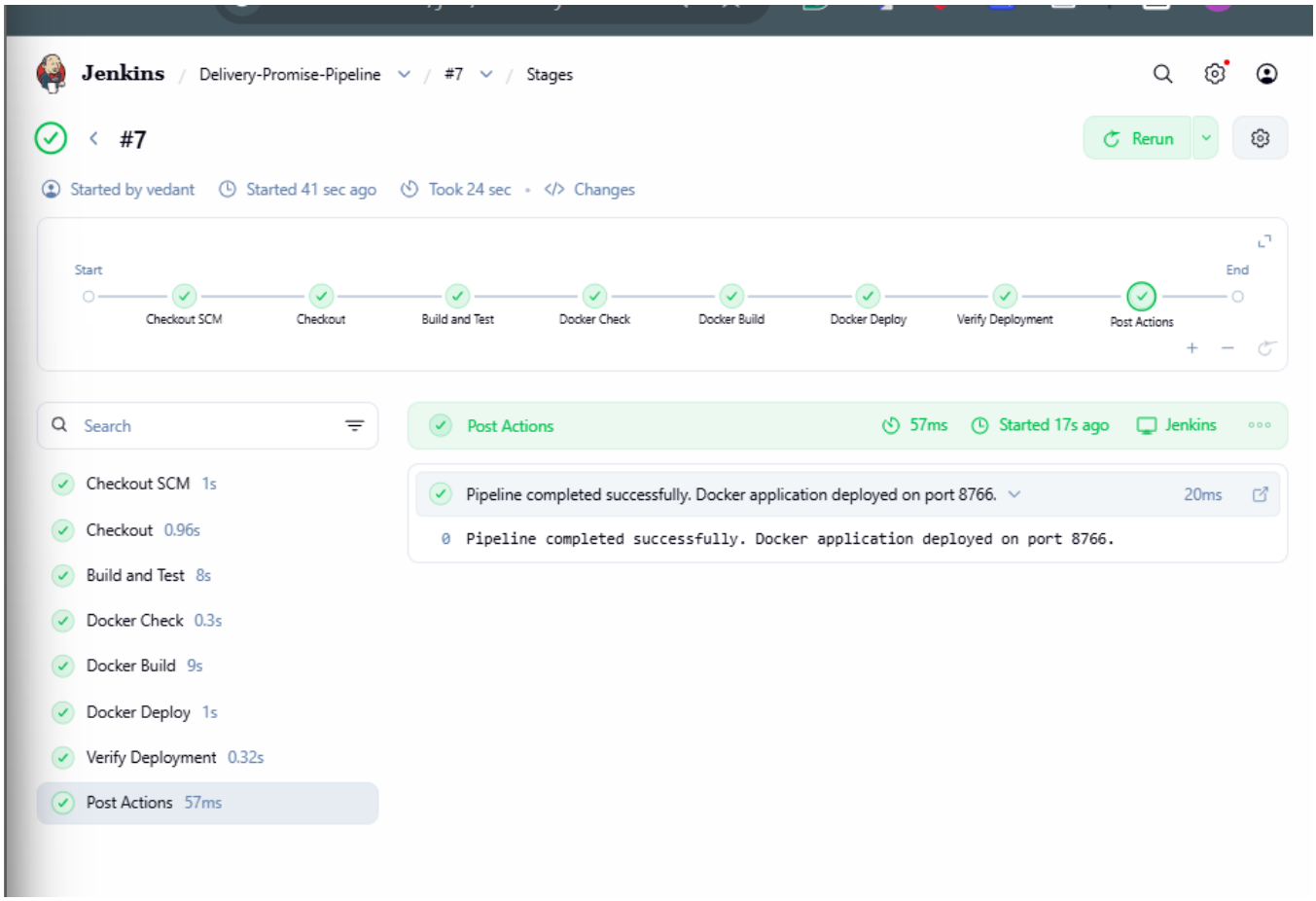
The first Docker build failed because the Dockerfile was not present in the Jenkins checkout of main. The Dockerfile was committed through the feature-branch/PR workflow and merged into main. The pipeline was then rerun.

4. Successful Pipeline

Build #7 passed Checkout SCM, Checkout, Build and Test, Docker Check, Docker Build, Docker Deploy, Verify Deployment, and Post Actions.

5. Evidence

Evidence: successful Jenkins-Docker deployment pipeline



6. Deployment Verification

The Docker application was accessed at http://localhost:8766/deliveries.

7. Result

Continuous deployment was achieved: Jenkins builds/tests the project, builds the image, replaces the container, starts the new container, and verifies the deployment.

Task 12 — Actual Project Screenshots

Screenshots captured during the implementation are included below as evidence. These are the actual project screenshots, not recreated diagrams.

Figure 1. Final Jenkins-Docker CD pipeline build #7: every stage passed and deployment was reported on port 8766.

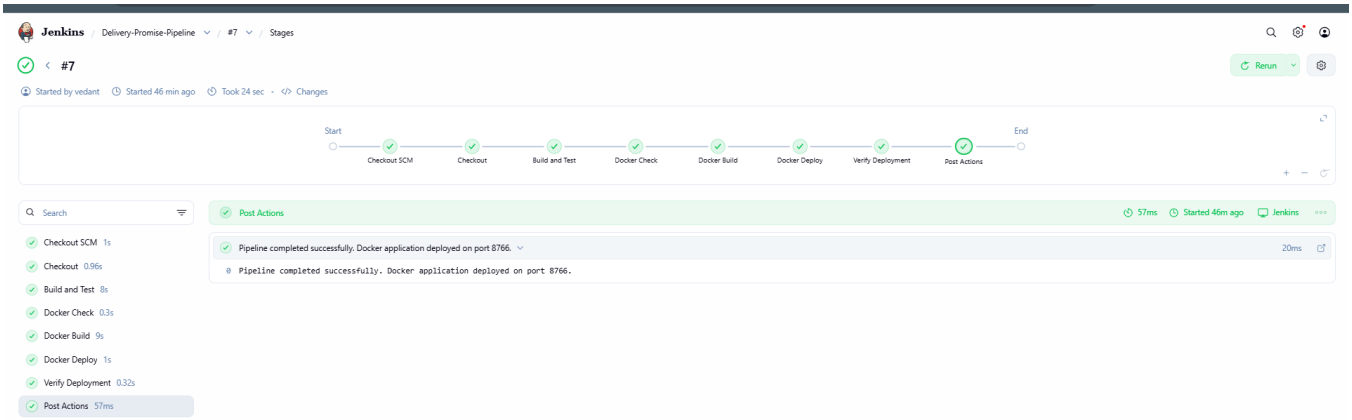


Figure 2. Browser verification of the containerized application at localhost:8766.

